

Abhinav Jagan Polimera

abhinavpolimera@gmail.com | LinkedIn | GitHub | Portfolio

U.S. Citizen | Authorized to work in the United States without sponsorship | Open to relocation

Education

Birla Institute of Technology and Science, Pilani - Hyderabad Campus

2020-2024

B.E. in Computer Science | CGPA: 8.2/10

Work Experience

Cisco Systems

Apr 2026 - Present

Software Developer 2 | Distributed Systems

On-site

- Built a semantic knowledge graph for the Cisco Silicon One router operating system, turning the codebase into structured, agent-readable context and improving LLM retrieval precision by approximately 30% while reducing token use.
- Co-developed an AI-driven infrastructure prototype that deployed 0.5-2 MB quantized ML models on Cisco routers at under 2% CPU; used large-scale telemetry to guide model selection and moved the work from concept to customer evaluation with a four-person core team.
- Designed and delivered ERSPAN over GUE for Microsoft, replacing GRE with UDP encapsulation; reduced tunnel overhead by 16%, avoided approximately 5 PB/year of telemetry traffic, and increased monitoring-fabric capacity by 2.3%.

Cisco Systems

Aug 2024 - Apr 2026

Software Developer | Distributed Systems

On-site

- Led egress traffic mirroring for Cisco 8000 (Silicon One), matching NCS 5000 behavior for SoftBank's migration and using component-level design plus AI-assisted development to reduce delivery time by 60%.
- Built AI-assisted developer tools, including a metrics dashboard used by 100+ engineers and a traffic-mirroring platform with constant-time support lookup, drag-and-drop router simulation, and MCP modules; cut issue reproduction time from five days to one.
- Owned delivery and debugging across SPAN, Lawful Intercept, and NetFlow workflows and led multi-platform topology bring-up across the Cisco router OS stack.

Cisco Systems

Jan 2024 - Jun 2024

Technical Intern | Distributed Systems

On-site

- Automated NTP, VRF, and OpenConfig validation in Python with 88% code coverage; redesigned and parallelized router-OS tests to cut execution time by 90%, earning a Cisco excellence award.

Shris Infotech

Jun 2022 - Jul 2022

Software Intern

Remote

- Built a face-recognition calling application in Python and Dart that reduced the average call flow from 10 clicks to 2.

Projects and Research

X-Agent - Local Multi-Agent Content Workflow

[Link](#)

- Built a local-first LangGraph and FastAPI application that transformed a consented persona interview into grounded drafts through retrieval, optional web research, consistency critique, and human approval.
- Implemented Ollama inference, local embeddings, Docker, and a browser UI; added automated tests, explicit approval before publication, and documented SSRF/DoS safeguards.

UAV-Based Suspicious-Activity Detection

[Link](#)

- Co-authored an IEEE paper on a Python/YOLO UAV surveillance system using object counts and activity duration to flag suspicious actions at remote locations.

NeRF and Photogrammetry for 3D Reconstruction

[Link](#)

- First-authored an ICAART 2024 comparison of NeRF and photogrammetry against ground-truth object dimensions; NeRF reached up to 99.89% accuracy on the evaluated geometric attributes.

Technical Skills

Programming: Python, C, C++, Shell

AI/ML: LLMs, agentic workflows, RAG, knowledge graphs, embeddings, LangGraph, MCP, model quantization, Ollama

Systems: Distributed systems, Linux, network operating systems, systems programming, routing, Cisco Silicon One, telemetry, packet processing, test automation, FastAPI, Docker

Computer Vision: OpenCV, YOLO, object detection, NeRF, photogrammetry

Interests and Activities

Photography; 3D modeling and printing; automotive design, F1, and sim racing; competitive chess (school team); karate (blue belt); Himalayan trekking and independent travel.